

REASONING SETTING

● OpenThinker3-7B vs parent

◆ Qwen3-8B, thinking on vs off

CORPORA (A, B)

■ Needs-reasoning corpora

■ No-reasoning corpora

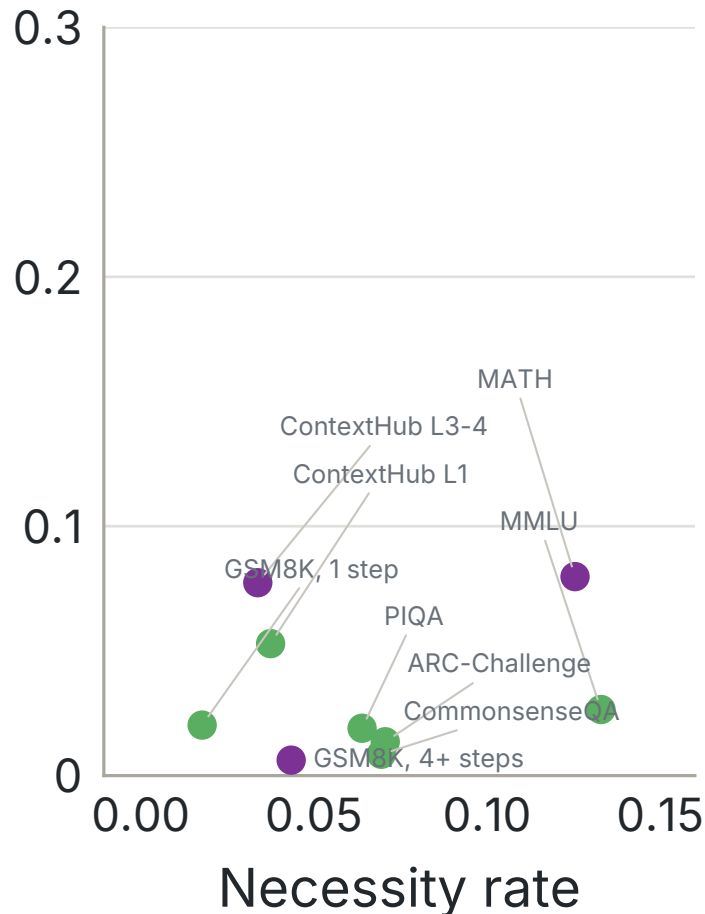
A · OPENTHINKER3-7B VS PARENT

B · QWEN3-8B, THINKING ON VS OFF

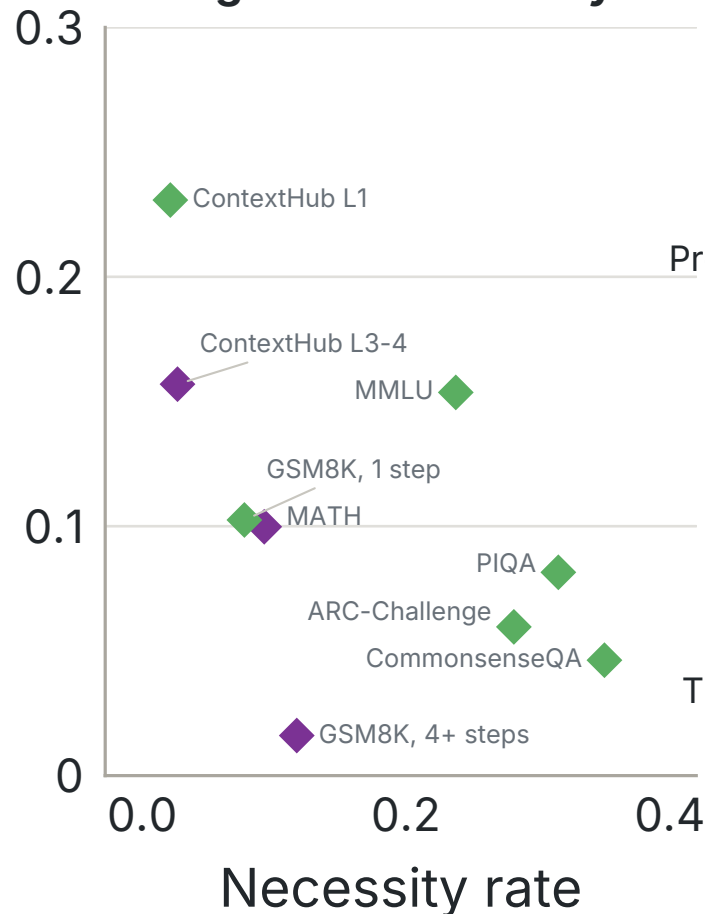
C · PER-QUESTION AUROC, 95% INTERVALS

End-of-thought gains are small
and do not track the necessity rate

Gain in own-answer acc@1
from end of thought ↗



Larger gains for the thinking toggle,
falling as the necessity rate rises



No map score identifies
which MATH questions need reasoning

